

ENSC 429: Digital Signal Processing**Course Project Information****Project Objectives**

This course project aims to assess the students' ability to use their knowledge of Digital Signal Processing in Engineering Design (ED).

This project is an **open-ended Engineering Design (ED)** project where students are expected to creatively develop engineering design candidates.

This project carries 25% of the course grade.

Project Description

The project topic is open-ended. Teams can choose to work on their own ideas, **but they must use some of the Digital Signal Processing concepts covered in this course, and must have strong ED components.** Students can choose to use any programming languages, such as Matlab, Python, or C/C++. Students can develop their projects based on open-source codes, but should have their own contributions.

Some examples of the project topics include:

- Speech signal processing
 - Speech denoising/enhancement
 - Speech recognition
 - Speaker separation
- Audio signal processing
 - Audio denoising/enhancement
 - Music classification
 - Musical instrument separation
- Image processing
 - Image denoising/enhancement
 - Image super-resolution
 - Image tone mapping: high dynamic range to low dynamic range
 - Image segmentation (semantic or instance segmentation)
 - Object recognition
- Video processing
 - Video denoising/enhancement
 - Video super-resolution (spatial and/or temporal)
 - Motion estimation
 - Video tone mapping: high dynamic range to low dynamic range
 - Video segmentation (semantic or instance segmentation)
 - Object tracking

Some useful resources are:

- Cloud GPU resources:
 - <https://gist.github.com/devinschumacher/87dd5b87234f2d0e5dba56503bfba533>
- IEEE Xplore: <https://ieeexplore.ieee.org/Xplore/home.jsp>
- The Calls for Papers of major conferences are good places to know the latest topics in the field
 - IEEE ICIP, ICME, CVPR, ICCV, ECCV
- <https://arxiv.org/> Many latest papers, especially on deep learning

Students are encouraged to adopt industry project management approach in their projects, such as the Agile Product Management:

- Incremental delivery
- Start from a MVP: Minimum Viable Product
- Integrate earlier: Don't wait till the last minute
- Fast iteration

Team Forming

This is a group project where student groups (each with 3-5 students, depending on the total enrollment) need to work together **professionally** and develop design candidates and finalize the design for the topic they choose.

It is recommended to choose your teammates as soon as possible. Students must note that there will be a peer evaluation at the end of the project. Free riding will not be tolerated and violators may lose the entire project mark.

Timeline

- Team forming: End of Week 3
- Proposal: End of Week 6
- Presentation: Week 13

Grading Policy

The project grade includes three parts:

- Proposal: 15%
- Presentation: 25%
- Final Report: 60%

1: Project Proposal

Teams need to submit a proposal in a short format described below to the instructional team and get their approval and feedback.

- Within 2 pages, including references (cover page not counted).
 - Single column, single space, 12-point font

- Introduction: problem statement, background, literature review (discuss the relevant papers)
- Objectives: what are you trying to accomplish, describe the engineering design, expected results
- Plan: How do you plan to achieve the objectives? Identify gaps in knowledge and resources, and how you plan to overcome these. Partition of the jobs among group members
- Tentative timeline: Identify milestones and deadlines to accomplish these milestones. Provide backup plan in case deadlines missed.
- References: List of relevant papers, datasets, etc.

2: Project Presentation

At the end of the semester, each team needs to give a presentation to the entire class. The length of each presentation is 7-20 minutes, depending on the class size. The presentation should include the following:

- Description of the project
- The results of the project
- Pre-recorded demo video: No on-site demo due to limited time

3: Project Final Report

Each group needs to submit a final report at the end of the semester. Students are expected to complete the project and document it in a professional format, similar to a scientific research paper.

The detailed requirements of the report are:

- Cover page: team number, group members, project title
 - No more than **20** pages, including references (cover page not counted)
 - Penalty for overlength: 20% per page
 - Single column, single space, 12-point font.
- Abstract: 100 words.
- Introduction: the problem, background, literature review
- Project description: the engineering design, implementations, results, discussions
- Member Contributions: Must have this section.
- Conclusion
- Submit both the report and the source codes via Canvas

Grading Rubrics

1) Proposal Rubric:

Criteria	A (85-100%) Excellent	B (73-84%) Good	C (56-72%) Satisfactory	D (50-55%) Needs Improvement	F (<50%) Unsatisfactory
Problem Statement (10%)	Clearly defined problem statement with relevance to course content.	Well-defined problem statement related to course content.	Adequate problem statement, but lacks specificity.	Vague or unclear problem statement, marginally related to course content.	Missing or irrelevant problem statement.
Objectives (20%)	Clear and specific project objectives	Well-defined objectives related	Objectives stated but lack	Unclear or incomplete	Missing objectives

	aligned with the problem statement, including clearly stated ED objectives.	to the problem, with reasonably defined ED objectives.	clarity or alignment, including the discussion of ED objectives.	objectives (including the ED objectives)	(including the ED objectives)
Plan (30%)	Clear plan on how to achieve objectives. Clearly identifies gaps in knowledge and resources, and plans to overcome these.	Plan to achieve objectives presented. Most of the gaps in knowledge and resources identified, with some idea on how to overcome these.	Vague or unclear plan provided. Some gaps in knowledge and resources identified, but unclear plan to overcome these.	Minimal plan to achieve objectives provided. Some gaps in knowledge or resources identified, but no plan on how to overcome these.	Missing plan on how to achieve objectives. Gaps in knowledge and resources not identified.
Tentative Timeline (20%)	Clearly identified milestones, clearly presented timeline in the form of a Gantt chart. Backup plans presented in case of milestones not achieved on time.	Milestones identified and timeline presented in the form of a Gantt chart. Some backup plans presented.	Some milestones identified, but timeline vague or unrealistic. Few, if any, backup plans.	Few, if any, milestones identified. Poorly presented timeline, no backup plans.	No milestones identified. No timeline presented. No backup plans.
References (20%)	All relevant references in complete IEEE format: Author names, paper title, venue, volume, pages, year.	Most relevant references provided and complete, but some are missing certain components, e.g., venue or year.	Some relevant references provided, but most are missing components, e.g., venue or year.	Few incomplete references provided.	No references provided.

2) Presentation Rubric

Criteria	A (85-100%) Excellent	B (73-84%) Good	C (56-72%) Satisfactory	D (50-55%) Needs Improvement	F (<50%) Unsatisfactory
Clarity (30%)	The presentation is very clear and informative, and the audience can	The presentation is mostly clear, offering some learning opportunities to the audience.	The presentation is somewhat clear, but with consistency issues, and little opportunity for	Difficult to follow, inconsistent, regurgitates well-known concepts, little to learn from.	Incoherent or incomprehensible presentation.

	learn something useful.		the audience to learn from it.		
Technical Strength (30%)	Technically strong and well-grounded, including ED components.	Technically solid, occasional unsupported claims or assumptions, including ED components.	Technically appropriate, occasional unsupported claims or assumptions, including ED components.	Technically borderline, many unsupported assumptions or claims, weak ED components.	Technically flawed, inaccurate, no ED components.
Delivery (20%)	Confident delivery with effective visuals and explanations.	Solid delivery, few issues with visuals or explanations.	Appropriate delivery, some issues with visuals and explanations.	Borderline delivery, many unclear visuals or explanations.	Poor delivery, no visuals, unclear explanations.
Time Management (10%)	Stays within allotted time, balances content appropriately, leaving time for Q&A.	Mostly adheres to time limits, but leaves little time for Q&A.	Some time management issues, leaves no time for Q&A.	Significant time overruns, unable to finish presentation within allotted time.	Grossly exceeds time limits, presentation has to be stopped.
Q&A (10%)	All questions are clearly answered, demonstrating mastery of the topic.	Questions are answered, demonstrating understanding of the topic.	Some questions are answered with hesitation or lack of clarity.	Most questions incorrectly answered, or not at all.	Unable to comprehend or answer questions.

3) Final Report Rubric

Criteria	A (85-100%) Excellent	B (73-84%) Good	C (56-72%) Satisfactory	D (50-55%) Needs Improvement	F(<50%) Unsatisfactory
Introduction and Literature Review (20%)	Clearly outlines the background, significance, and the state of the art on the topic, supported by high-quality references. Clearly states the motivations and objectives of the project.	Outlines the background, significance, and the state of the art on the topic, supported by references. States the motivations and objectives of the project.	Some background on the topic with a few references. Motivations and objectives of the project are vague or partially unclear.	Missing background on the topic and/or references. Motivations and objectives of the project are unclear.	No background on the topic. Missing motivation and objectives of the project.

Description of the Theory, Design and Implementation (30%)	The relevant theory is described very clearly using appropriate equations, terminology, figures, etc. Material used from other sources is cited very clearly. Final design is clearly a result of meaningful, creative, well-described design iterations . Clearly identifies added value, e.g., improving an algorithm, developing a new model, etc.	The relevant theory is described clearly. Material used from other sources is cited. Final design is a result of meaningful design iterations . Identifies added value, e.g., improving an algorithm, developing a new model, etc.	The relevant theory is described adequately. Material used from other sources is usually cited. Final design is a result of design iterations . Some added value is identified, e.g., improving an algorithm, developing a new model, etc.	The relevant theory lacks proper description. Material used from other sources is often not cited. Final design is a result of trivial design iterations . Little added value is identified.	The relevant theory is missing. Material used from other sources is not cited. Final design is not a result of design iterations . Added value not clear or non-existent.
Results, Analyses, and Conclusions (30%)	Comprehensive, meaningful results, verified by comparing with other sources. The results are clearly presented, analyzed and discussed in a logical and insightful way, and related to ED components.	Solid results, with some comparison with other sources. The results are analyzed and discussed in a logical way, and related to ED components.	Adequate results, with little or no comparison with other sources. The results are analyzed and discussed, but connection to ED components is unclear.	Weak results, no comparison with other sources. Superficial discussion, if any. Lacks critical analysis, including the relationship to ED components.	No results obtained. Missing discussion, including the ED components.
Reflections on the Plan from the Proposal (10%)	Clear, thoughtful reflections on the plan: what worked, what didn't, and why? Clearly explains what was done to adjust the plan, if needed,	Solid reflections on the plan: what worked, what didn't, and why? Explains what was done to adjust the plan, if	Adequate reflections on the plan: what worked, what didn't, and why? Some explanations on what was done to adjust the	Minimal reflections on the plan. Lacks explanation of what was done to adjust the plan, if needed, during the project. Minimal	No reflections on the plan. No summary of the lessons learned.

	during the project. Clear summary of the lessons learned.	needed, during the project. Solid summary of the lessons learned.	plan, if needed, during the project. Adequate summary of the lessons learned.	summary of the lessons learned.	
Discussion of Individual Contributions (10%)	Clear descriptions of individual contributions of group members.	Reasonable descriptions of individual contributions of group members.	Brief or vague descriptions of individual contributions of group members.	Incomplete descriptions of individual contributions of group members.	No descriptions of individual contributions of group members.
Overall Writing Quality (10%)	Writing is very clear, well balanced, and logical, with smooth transitions. No grammatical errors. Figures, tables, etc. are properly labeled, clear, visually high quality, and clearly discussed / interpreted in the text.	Writing is clear, balanced, and logical. Few, if any, grammatical errors. Figures, tables, etc. are labeled, clear, visually good quality, and discussed / interpreted in the text.	Writing is mostly clear, balanced, and mostly logical. Some grammatical errors. Figures, tables, etc. are labeled and usually discussed or interpreted in the text.	Writing is mostly unclear, imbalanced, and lacks logical coherence. Considerable number of grammatical errors. Figures, tables, etc. are rarely labeled and usually not discussed or interpreted in the text.	Writing is unclear, imbalanced, without logical coherence. Many grammatical errors. Figures, tables, etc. are missing, or are unlabeled and not discussed or interpreted in the text.
References (10%)	Properly formatted references in the IEEE format. The references are cited properly throughout the report.	Mostly correct formatting and citing with minor errors.	Some formatting issues or inconsistent citations.	Significant formatting errors or missing citations.	No references or incorrect format.